

Question

In a specially designed sealed system, $[(n - C_4H_9)_4N][H_2PO_4]$ reacts with $HSiCl_3$ for 6 days, yielding colorless needle-shaped compound **A** and releasing gas, with a total of three products (Reaction 1). Further analysis reveals that **A** is a 1 : 1 type salt with a molar mass of 542.3 g/mol. The anion of **A** contains only three elements: P, Si, Cl; atoms of the same element have the same chemical environment.

A is sensitive to water. 1 mol **A** decomposes upon contact with water to produce 1 mol of a highly toxic gas (Reaction 2).

At 382 K, **A** reacts fully with $Ph(CH_2)_4Cl$ followed by water treatment, yielding a disubstituted phosphine (Reaction 3).

Which of the following statements is correct:

- A.** All other options are incorrect
- B.** The anion of **A** contains a tetrahedral structure.
- C.** The chemical equation for reaction 1, balanced with the simplest coefficient ratio, has the sum of the product coefficients equal to the sum of the reactant coefficients.
- D.** The sum of the coefficients of the products, without Si, in the balanced chemical equation for reaction 2, using the simplest integer ratio, is a composite number.
- E.** The disubstituted phosphine product of reaction 3 contains fewer than 40 atoms.

Answer

Correct Answer: **C**

Detailed Explanation

The cation of **A** is obviously $[(n - C_4H_9)_4N]^+$; since it is a 1 : 1 type salt, the chemical formula of the anion can be set as $P_xSi_yCl_z$.

CHECKPOINT

1 PTS

The cation of **A** is $[(n - C_4H_9)_4N]^+$

The relative molecular mass of the anion is $542.3 - (14 + (12 \times 4 + 9) \times 4) = 300.3$. Considering that the same atoms have the same chemical environment, z is likely to be an even number. First assume $x = y = 1$, then $z = (300.3 - 31 - 28.1)/35.45 = 6.8$, which is not an integer.

Let $x = 1, y = 2, z = (300.3 - 31 - 28.1 \times 2)/35.45 = 6$, which is an integer. Observing the valence, Cl is obviously -1, at this time Si is +4, P is -3, which are all reasonable valences. Therefore, the anion of **A** is $PSi_2Cl_6^-$, and the chemical formula is $[(n - C_4H_9)_4N][PSi_2Cl_6]$.

CHECKPOINT

1 PTS

Based on the molecular weight, the anion of **A** is deduced to be $PSi_2Cl_6^-$

CHECKPOINT

1 PTS

The chemical formula of **A** is $[(n - C_4H_9)_4N][PSi_2Cl_6]$

The structure of $PSi_2Cl_6^-$ is obviously that two $SiCl_3$ units are bridged by a P atom, which is a connection of two distorted tetrahedra at a common point. There is no regular tetrahedron, so option B is incorrect.

CHECKPOINT

1 PTS

The structure of $PSi_2Cl_6^-$ is that two $SiCl_3$ units are bridged by a P atom, which is a connection of two distorted tetrahedra at a common point

The reaction of $[(n - C_4H_9)_4N][H_2PO_4]$ with $HSiCl_3$ is obviously the reduction of high-valent P by the Si – H bond, and the gas released should be hydrogen; since it is mentioned later that **A** is unstable to water, this reaction must not produce water; therefore, only $(SiCl_3)_2O$ can be produced. The balanced reaction equation is:

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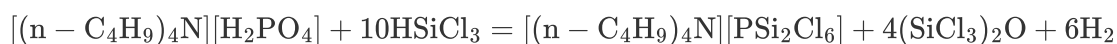
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Reaction 1 is the reduction of high-valent P by the Si – H bond, releasing hydrogen

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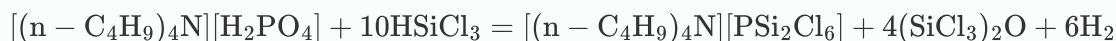
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Reaction 1 must not produce water; $(SiCl_3)_2O$ is produced



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1 PTS



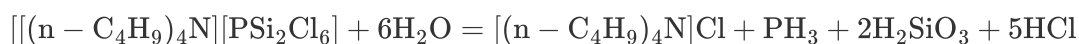
The sum of the coefficients of the reactants and the sum of the coefficients of the products are both 11, so option C is correct.

The reaction of $[(n - C_4H_9)_4N][PSi_2Cl_6]$ with water is similar to a hydrolysis reaction, which is not a redox reaction, so the highly toxic gas produced should be PH_3 . Si(IV) should be converted into silicic acid, and Cl is converted into HCl. Thus, balance the equation for reaction 2:

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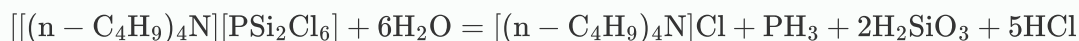
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The highly toxic gas is PH_3



CHECKPOINT

1 PTS



The sum of the coefficients of the products without Si is 7, which is a prime number, so option D is incorrect.

CHECKPOINT

1 PTS

The sum of the coefficients of the products without Si in reaction 2 is 7

The reaction of $[(n - C_4H_9)_4N][PSi_2Cl_6]$ with $Ph(CH_2)_4Cl$ can be regarded as the nucleophilic attack of the lone pair electrons of P on the haloalkane, so the chemical formula of the obtained disubstituted phosphine should be $[Ph(CH_2)_4]_2PH$, which contains 48 atoms, so option E is incorrect.

CHECKPOINT

1 PTS

The chemical formula of the disubstituted phosphine is $[Ph(CH_2)_4]_2PH$